IV Answer the following questions

1. Write in brief about the central dogma of molecular biology.
2. Describe the Meselson-Stahl experiment and the conformation of semi-conservative model.
3. Explain the events involved in DNA Replication briefly.
4. Compare the different models of DNA replication.
5. Differentiate between the process of DNA replication in prokaryotes and eukaryotes.
6. Explain the role of the DNA polymerase in DNA replication.
7. Describe the steps involved in PCR amplification.
8. Give an Overview of transcription.
9. Compare Prokaryotes and Eukaryotes with respect to Transcription
10. Write briefly about the Molecular events in translation.
11. Write short notes on Recombinant DNA technology.
12. Explain the Sanger sequencing procedure in detail.
13. Discuss about the Post transcriptional modification and Post-translational modifications events.
14. Compare the three main types of RNA in living organisms.
15. Discuss about Next Generation Sequencing (NGS) Technologies and their role in genomics.
16. Explain the different applications of NGS technologies in genomics.

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